Project Report: Uni-Network v4

Topology Expansion & Advanced Layer 2 Security

1. Phase 4 Overview

Following the successful implementation of core routing and fundamental security measures in earlier phases, Phase 4 of the "Uni-Network" project focuses on scaling the campus architecture and mitigating advanced local-area network threats. This phase introduces two new specialized academic wings—an IoT Research Lab and a Cybersecurity Special Interest Group (SIG) "Subnet"—reflecting real-world university infrastructure expansion. Furthermore, the network was hardened against Man-in-the-Middle (MitM) attacks using DHCP Snooping, and a Centralized Syslog Server was deployed to provide real-time, campus-wide security monitoring.

2. Network Topology Expansion

To simulate organizational growth, the physical and logical topology was expanded to accommodate two new departments, requiring new IP address spaces and dynamic routing updates.

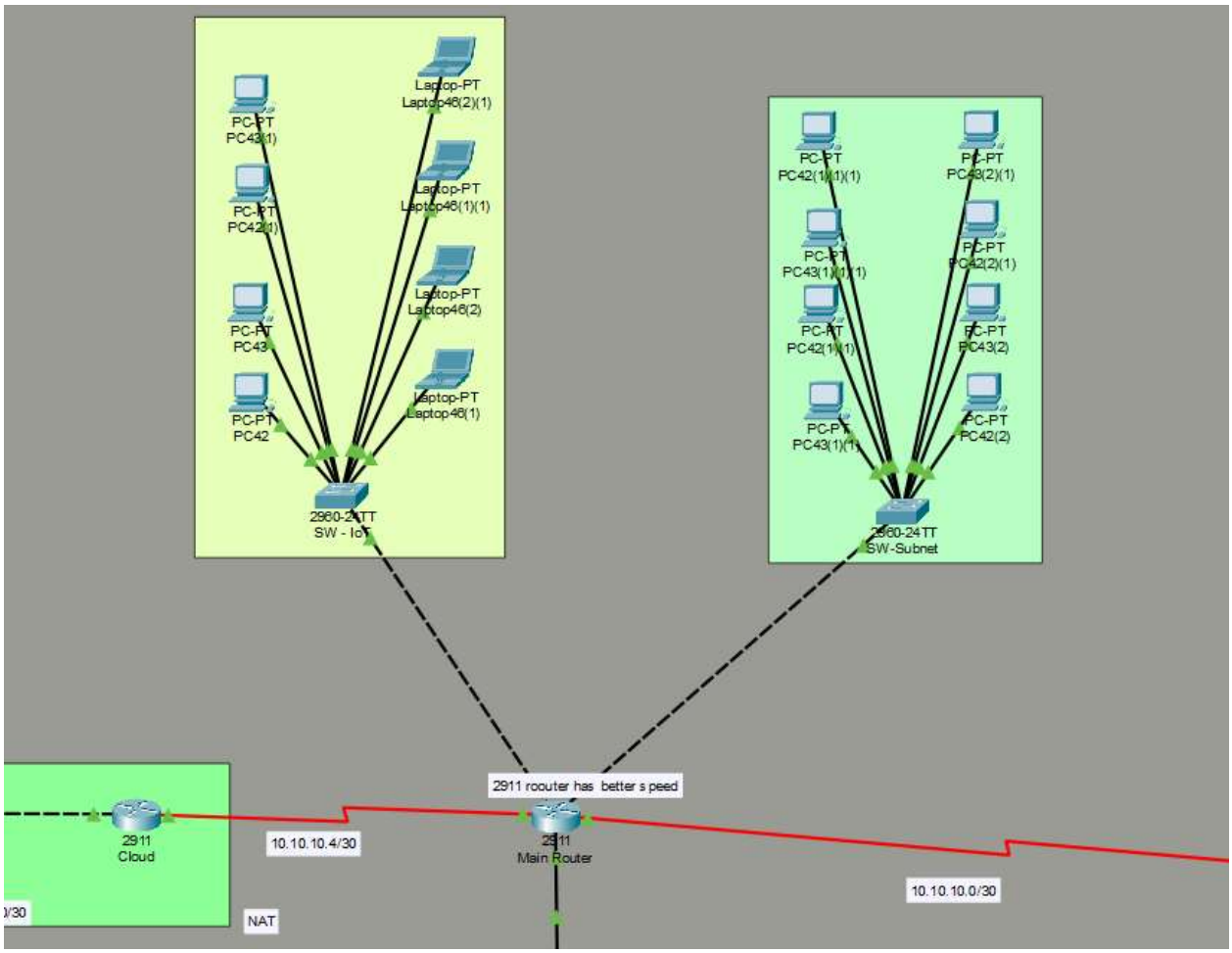
2.1 IP Addressing and Subnetting

Two new Class C subnets were introduced to the existing architecture:

- **IoT Research Lab:** 192.168.30.0 /24
- **Cybersecurity "Subnet":** 192.168.40.0 /24

2.2 Physical Integration and DHCP Allocation

- **Implementation:** Dedicated Cisco 2960 switches (SW-IoT and SW-Subnet) were deployed and uplinked to the core routing infrastructure via GigabitEthernet interfaces.
- **Dynamic Addressing:** To facilitate automated IP distribution for these new labs, dedicated DHCP pools (IOT_LAB and SUBNET_LAB) were configured on the Core Router, setting the respective default gateways to 192.168.30.1 and 192.168.40.1.
- **Routing Updates:** The RIP version 2 routing protocol was updated to advertise the 192.168.30.0 and 192.168.40.0 networks, ensuring seamless inter-VLAN communication across the newly expanded campus.



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3. Advanced Layer 2 Security (DHCP Snooping)

As network access expands, the risk of internal Layer 2 attacks increases. A critical vulnerability is the introduction of a rogue DHCP server, where an unauthorized device assigns fraudulent IP addresses to endpoints, routing their traffic through an attacker's machine (MitM attack).

3.1 Mitigating Rogue DHCP Servers

- **Implementation:** DHCP Snooping was globally enabled on the new distribution switches (ip dhcp snooping) and actively applied to the operational VLANs.
- **Port Trust Assignments:** The switch ports acting as uplinks to the Core Router were explicitly configured as trust ports. All other edge ports connecting to end-user PCs remained untrusted by default. If a rogue router attempts to issue DHCP OFFER packets from an untrusted port, the switch instantly drops the traffic.

- **Rate Limiting:** To prevent DHCP starvation attacks—where an attacker floods the network with IP requests to exhaust the DHCP pool—rate limiting was applied to the access ports, restricting them to 10 DHCP packets per second.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#ip dhcp snooping
Switch(config)#ip dhcp snooping vlan 1
Switch(config)#interface GigabitEthernet0/2
Switch(config-if)#ip dhcp snooping trust
Switch(config-if)#exit
Switch(config)#interface range FastEthernet0/1-24
Switch(config-if-range)#ip dhcp snooping limit rate 10
Switch(config-if-range)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#write memory
Building configuration...
[OK]
Switch#
```

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4. Centralized Security Monitoring (Syslog)

Prior to Phase 3, security mechanisms like Port Security and ACLs actively dropped unauthorized traffic, but these events were only logged locally on individual devices. This created a fragmented monitoring environment.

4.1 Syslog Server Deployment

- **Implementation:** A dedicated Syslog Server was deployed within the secure Administration Network (192.168.20.100).
- **Device Configuration:** All core routers and distribution switches were configured to forward their system logs to this centralized IP address (logging 192.168.20.100).
- **Timestamping:** To ensure accurate forensic timelines during a security event, the service timestamps log datetime msec command was applied globally. This forces every network device to attach exact dates and millisecond-accurate timestamps to every generated alert.

```
R1(config)#interface GigabitEthernet0/2
R1(config-if)#ip address 192.168.40.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#ip dhcp pool IOT_LAB
R1(dhcp-config)#network 192.168.30.0 255.255.255.0
R1(dhcp-config)#default-router 192.168.30.1
R1(dhcp-config)#exit
R1(config)#ip dhcp pool SUBNET_LAB
R1(dhcp-config)#network 192.168.40.0 255.255.255.0
R1(dhcp-config)#default-router 192.168.40.1
R1(dhcp-config)#exit
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#network 192.168.30.0
R1(config-router)#network 192.168.40.0
R1(config-router)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#write memory
Building configuration...
[OK]
R1#%DHCPD-4-PING_CONFLICT: DHCP address conflict:  server pinged 192.168.40.1.

R1#
```

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5. Testing and Verification

To validate the Phase 3 implementations, active penetration tests and connectivity checks were performed within the Cisco Packet Tracer environment.

- **Expansion Verification:** End devices in the new IoT and Cybersecurity labs successfully leased IP addresses from the 192.168.30.0 and 192.168.40.0 pools. Pings from these labs to the original Student Subnet were successful, verifying the RIP routing updates.
- **Active Threat Simulation (Syslog & Port Security):** A rogue laptop was intentionally connected to a secured port on the SW-IoT switch.
  - *Result:* The switch immediately transitioned the port to an err-disable (shutdown) state.
  - *Monitoring Result:* The event was instantly transmitted to the Centralized Syslog Server. The server dashboard successfully recorded a **Severity 2 (Critical)** warning, displaying the exact timestamp, the unauthorized MAC address, and the compromised switch interface.

```
R1>enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#logging 192.168.20.100
R1(config)#service timestamps log datetime msec
R1(config)#end
R1#
*Mar 02, 02:24:49.2424: SYS-5-CONFIG_I: Configured from console by console
*Mar 02, 02:24:49.2424:  %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.20.100 port 514
started - CLI initiated
R1#write memory
Building configuration...
[OK]
R1#
```

6. Conclusion

The completion of Phase 3 transforms the Uni-Network from a static, isolated topology into a dynamic, actively monitored environment. By mitigating Layer 2 vulnerabilities through DHCP Snooping and establishing a centralized incident response dashboard via Syslog, the campus network now adheres to stringent enterprise security standards, ready to handle real-world operational threats.